

CAIRNS AAF, AL, USA

WMO: 722269

Lat: 31.267N

Lon: 85.717W

Elev: 301

StdP: 14.54

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.6	29.7	10.1	9.3	34.5	15.7	12.3	38.3	20.6	54.0	18.9	53.2	6.1	320	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	95.7	75.9	93.4	76.1	91.5	75.7	80.4	88.7	79.3	87.8	78.3	86.5	7.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	150.9	83.4	77.2	143.3	82.2	76.4	139.3	81.6	44.2	88.9	43.0	87.8	41.9	87.6	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.6	12.9	19.7	98.3	4.2	2.4	16.7	100.1	14.2	101.5	11.9	102.9	8.8	104.6		
(5)			17.8	82.5	4.2	1.9	14.8	83.9	12.3	85.0	9.9	86.1	6.9	87.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	66.9	49.8	53.8	59.8	66.1	74.3	79.7	81.5	80.6	77.0	67.9	58.2	52.9	(6)
(7)		DBStd	13.26	10.28	9.51	8.44	6.35	5.46	3.83	2.86	3.00	4.85	7.33	8.15	9.54	(7)
(8)		HDD50	317	133	63	21	0	0	0	0	0	0	1	20	79	(8)
(9)		HDD65	1758	478	328	199	63	8	0	0	0	3	56	234	389	(9)
(10)		CDD50	6476	128	169	325	484	752	892	977	948	809	557	265	170	(10)
(11)		CDD65	2442	8	14	38	97	295	442	512	483	362	147	29	15	(11)
(12)		CDH74	22164	23	71	305	892	2748	4197	4946	4478	3097	1187	169	51	(12)
(13)		CDH80	9324	0	4	38	210	1118	1976	2311	2011	1302	334	18	2	(13)
(14)	Wind	WSAvg	6.1	6.9	7.1	7.1	6.6	6.0	5.4	5.0	4.9	5.7	6.0	6.4	(14)	
(15)	Precipitation	PrecAvg	55.4	5.3	4.6	5.6	4.5	3.0	5.5	6.4	4.8	4.3	3.1	4.4	4.0	(15)
(16)		PrecMax	70.6	13.8	9.4	11.0	9.4	8.9	11.9	20.2	8.2	16.9	8.0	12.0	9.3	(16)
(17)		PrecMin	36.4	1.3	1.2	0.4	1.3	0.2	2.1	2.5	2.3	0.7	0.5	1.3	1.5	(17)
(18)		PrecStd	10.1	2.8	2.3	2.9	2.3	2.0	3.0	3.9	1.7	3.7	2.0	2.6	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	79.6	84.0	87.8	95.0	99.1	98.8	97.0	96.1	90.0	82.1	79.0	(19)
(20)			MCWB	65.8	66.9	66.1	69.4	72.4	75.3	77.0	76.3	73.9	73.1	69.6	69.3	(20)
(21)		2%	DB	72.9	76.4	80.9	84.8	91.3	95.5	95.0	93.9	92.5	86.5	78.7	74.9	(21)
(22)			MCWB	64.3	65.4	65.0	68.6	71.6	75.6	77.5	77.2	73.8	71.2	67.1	67.1	(22)
(23)		5%	DB	69.9	72.3	77.4	82.2	89.1	93.0	92.8	91.7	90.1	83.8	75.3	71.7	(23)
(24)			MCWB	63.7	63.3	63.7	66.9	71.4	75.7	77.3	76.9	74.3	70.1	65.6	66.3	(24)
(25)		10%	DB	66.1	69.2	74.3	79.5	86.4	90.3	90.7	90.0	87.7	81.0	72.4	68.2	(25)
(26)			MCWB	61.4	62.2	62.3	65.1	70.6	75.1	76.9	76.3	73.8	68.2	64.4	64.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.4	70.2	71.4	74.2	76.6	80.4	82.1	82.4	79.0	77.6	74.0	72.6	(27)
(28)			MCDWB	71.8	74.8	77.0	80.6	85.7	89.9	90.6	90.0	86.0	83.3	77.1	75.6	(28)
(29)		2%	WB	67.4	68.3	69.2	72.2	75.3	79.0	80.5	80.5	77.8	75.2	71.4	69.5	(29)
(30)			MCDWB	70.9	72.7	74.1	79.0	83.7	88.7	89.3	88.0	85.5	80.4	74.7	72.2	(30)
(31)		5%	WB	64.9	66.3	67.4	70.4	74.2	78.0	79.4	79.2	76.8	73.4	68.9	67.3	(31)
(32)			MCDWB	68.4	70.8	73.4	77.2	82.6	87.5	88.3	86.8	84.5	79.3	73.1	70.3	(32)
(33)		10%	WB	62.3	63.7	65.5	68.7	73.1	77.0	78.4	78.1	75.7	71.7	66.2	64.7	(33)
(34)			MCDWB	65.8	67.2	71.4	75.2	81.5	86.0	86.7	85.5	83.0	77.6	71.0	67.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.2	21.7	22.9	23.4	22.2	19.6	18.7	18.6	19.5	22.1	23.0	20.0	(35)
(36)			MCDBR	21.4	22.9	24.7	25.3	24.4	23.4	21.4	21.2	22.5	23.3	23.4	20.6	(36)
(37)			MCWBR	14.0	13.6	12.0	10.9	8.2	7.1	6.9	7.3	7.7	9.9	13.6	14.0	(37)
(38)		5% WB	MCDWB	18.8	19.0	20.0	19.9	20.4	20.2	19.0	18.1	18.0	17.7	19.1	18.1	(38)
(39)			MCWBR	14.5	13.6	11.9	10.8	8.2	7.5	7.5	7.5	7.2	9.2	13.5	14.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.324	0.333	0.356	0.381	0.433	0.469	0.513	0.501	0.431	0.372	0.345	0.330	(40)
(41)		taud		2.521	2.497	2.433	2.368	2.256	2.215	2.104	2.136	2.310	2.449	2.470	2.519	(41)
(42)		Ebn at Noon		287	295	294	289	273	262	250	252	266	277	276	279	(42)
(43)		Edh at Noon		27	31	35	39	44	46	51	49	39	32	29	26	(43)
(44)	All-Sky Solar Radiation	RadAvg		950	1135	1520	1859	2034	1934	1880	1749	1585	1379	1070	826	(44)
(45)		RadStd		60	142	124	153	141	151	117	116	118	151	91	88	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-167	+299	+196

Nomenclature: See separate page